FINAL YEAR PROJECT / DEPARTMENT OF COMPUTER SCIENCE

ASL Communication System

Bridging communication between sign language users and non-signing individuals



BIDIRECTIONAL

Sign ↔ Text + Speech
Text ↔ 3D Avatar

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What we will cover today

01
Introduction & Motivation
Why this problem matters

02
Problem Statement
The mutual communication barrier

03
Objectives
Clear goals of the project

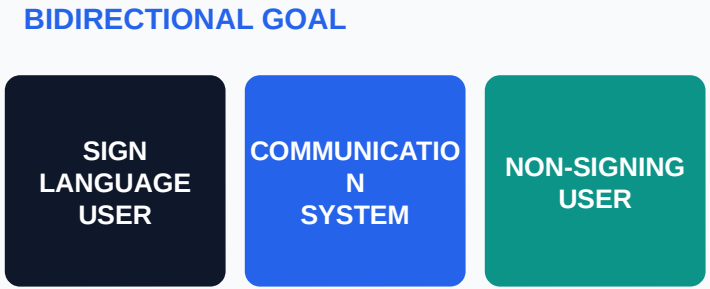
04
System Design
Flow, Architecture & Technology

05
Implementation Approach
How the system is built

06
Differentiation & Future
Unique value and next steps

A shared conversation needs a shared interface.

Communication becomes difficult when people use different methods. Sign language is essential for many Deaf and hard-of-hearing people, yet non-signers may not understand it. Computer vision, machine learning, web applications and 3D animation can help bridge this gap.



Sign Language Users
Need natural, real-time communication without constant interpreter dependency.

Non-Signing Individuals
Need an accessible way to understand and respond to sign language.

Technology Bridge
CV + ML + Web + 3D animation create a shared communication interface.

The barrier is mutual — not one-sided.

SIGN-LANGUAGE USERS

- Interpreter dependency in daily life
- Barriers in education, healthcare & services
- Typing interrupts natural conversation flow
- Limited real-time communication options

NON-SIGNING INDIVIDUALS

- Cannot interpret signs without training
- Cannot respond visually in real time
- Interaction becomes slow and awkward
- Missed opportunities for inclusion

Clear goals that guide the prototype.

01

Real-time Sign Recognition

Detect and classify selected ASL signs from camera input using hand landmarks and sequential models (LSTM / GRU).

02

Sign → Text + Speech

Convert recognized signs into readable text and natural speech output for non-signing users.

03

Text → 3D Sign Animation

Translate typed text into corresponding 3D avatar sign animations for visual response.

04

Integrated Web Application

Deliver both pathways in a single browser-based interface with camera, text I/O and 3D visualization.

05

Model Comparison

Train and evaluate LSTM vs GRU on the same dataset to compare accuracy, speed and suitability.

06

Lightweight Pipeline

Use MediaPipe landmarks instead of full video frames to keep the system efficient and practical.

Two pathways meet in one bidirectional experience.

SIGN → TEXT + SPEECH

PERFORM SIGN → CAMERA → PROCESS → RECOGNIZE → TEXT → SPEECH

TEXT → 3D SIGN

ENTER TEXT → CHECK VOCAB → FIND SIGN → SELECT ANIM → 3D AVATAR → UNDERSTAND

A modular stack from browser to model inference.

EXPERIENCE
React UI • Camera • Text I/O

VISUALIZATION
Three.js Avatar • Blender Animations

API & SERVICES
FastAPI • Python TTS

VISION & DATA
OpenCV • MediaPipe • NumPy

ML & EVALUATION
TensorFlow / Keras • LSTM / GRU • Scikit-learn

Choices that balance fit, learning curve and integration.

Frontend

React
Component-based UI & camera integration

Three.js
Real-time 3D avatar rendering

Backend & Services

FastAPI
High-performance Python API

Python TTS
Text-to-speech output

Computer Vision

MediaPipe
Reliable hand landmark detection

OpenCV
Camera & image processing

Machine Learning

TensorFlow / Keras
LSTM & GRU model training

Scikit-learn
Baselines & evaluation metrics

Assistive technology with practical and research value.



01

INDIVIDUAL

Supports direct interaction and reduces dependency on typing or interpreters.



02

EDUCATION

Demonstrates accessible classroom possibilities for inclusive learning.



03

PUBLIC

Potential use at desks, offices, service counters and public facilities.



04

RESEARCH

Compares sequential models and integrates AI, web and 3D in one system.

Distinct through integration, experimentation and restraint.

01

BIDIRECTIONAL

Sign → text → speech and text → 3D sign in one continuous experience.

02

COMPARATIVE

LSTM vs GRU trained and evaluated on the same dataset with clear metrics.

03

LIGHTWEIGHT

Landmark + movement-sequence recognition instead of heavy full-frame models.

04

INTEGRATED

AI, text, speech and 3D visualization delivered together in a single web app.

A clear path from prototype to expressive platform.

01 PROTOTYPE

- 5–8 Selected ASL Signs
- Real-time Camera Input
- Hand Landmark Detection
- LSTM & GRU Recognition
- Sign → Text + Speech
- Text → 3D Avatar
- Web-based Interface

02 EXPANDED

- Expanded ASL Vocabulary
- Optimized ML Models
- Advanced Hand Tracking
- Full Bidirectional Flow
- Performance Evaluation
- Complete Web Application
- User Testing

03 ADVANCED

- Grammar & Expression
- Head & Body Motion
- Multiple Sign Languages
- Continuous Signing
- Mobile Support
- Cloud Deployment
- Real-world Pilots

From barrier to exchange.

This prototype compares LSTM and GRU recognition models, then integrates selected-vocabulary signing with text, speech and 3D visual output in one assistive web application.

SIGN ↔ AI SYSTEM ↔ TEXT • SPEECH • 3D SIGN

A shared interface for shared meaning.

THANK YOU

Questions & Discussion

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